Advanced Biotechnology

Lorenzo Di Michele
Professor of Bionanoscience
CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecturer Information
Lorenzo Di Michele

Research: Nucleic acid nanotechnology, bottom-up synthetic biology, biophysics. See www.dimichelelab.org/

Email: ld389@cam.ac.uk (preferred, expect response in 2 working days)

CEB Office: Room 3.13

Course Information

- 16 Lectures
- 4 example papers and 4 supervisions
- All slides and example papers to be posted on Moodle

Useful resources

General References

- Voet, D. & Voet, J. *Biochemistry*. 4th ed. Hoboken, NJ: Wiley, 2011
- Alberts, B., Johnson, A., Lewis, J., et al. *Molecular Biology of the Cell*. 6th ed. New York: Garland Science, 2015

Nucleic Acids

- Blackburn, G.M., Gait, M.J., Loakes, D., & Williams, D.M. Nucleic Acids in Chemistry and Biology. 3rd ed. Cambridge: RSC Publishing, 2006.

Proteins & Bioprocessing

- Clark, D.P. *Biotechnology*. 2nd ed. Amsterdam: Academic Press, 2016.
- Cytiva. *Principles and Methodology Handbooks*. Web resources, 2023 https://www.cytivalifesciences.com/en/us/support/handbooks
- Doran, P.M. *Bioprocess Engineering Principles*. 2nd ed. London: Academic Press, 2013

Course Information

Course builds upon material covered by the CEBT IA and IB courses on Cell Biology and Biotechnology Fundamentals, but also basic chemistry and thermodynamics.

It is divided in **three parts**

1) Molecular biotechnology methods

- Nucleic acids chemistry, synthesis, sequencing
- Nucleic acid nanotechnology
- Protein engineering, de novo design and applications

2) Principles of industrial bioprocessing and bioseparations

- Overview of industrial bioprocessing
- Focus on bioseparations and protein purification

3) Synthetic biology

- Synthetic biology and gene circuit design
- Cell free systems and synthetic cells

Course outline (tentative)

1) Molecular Methods in Biotechnology

Lecture 1: Intro + Nucleic acids structure, thermodynamics and kinetics

Lecture 2: Nucleic Acids Synthesis and Assembly

Lecture 3: Nucleic Acid Sequencing I

Lecture 4: Nucleic Acid Sequencing II

Lecture 5: Quantitative PCR and nucleic acid aptamers

Lecture 6: DNA nanotechnology

Lecture 7: Review of protein structure and folding (Prof. Graham Christie)

Lecture 8: Protein engineering and de-novo design (Prof. Mark Howarth)

Course outline (tentative)

2) Bioprocessing

Lecture 9: Bioprocessing overview + Cell removal I

Lecture 10: Downstream bioprocessing topics: Cell removal II + Cell disruption

Lecture 11: Downstream bioprocessing topics: Primary isolation I

Lecture 12: Downstream bioprocessing topics: Primary isolation II + Chromatography

3) Synthetic Biology

Lecture 13: Gene expression and gene regulation (Prof. Somenath Bakshi)

Lecture 14: Synthetic circuit design and characterization (Prof. Somenath Bakshi)

Lecture 15: Cell free synthetic biology and Synthetic Cells

Lecture 16: Spare lecture or revision

Lecture 1

Nucleic acids structure, thermodynamics and kinetics



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Lecture outline

1) Chemistry and structure of DNA and RNA
2) Hybridisation thermodynamics
3) Hybridisation kinetics

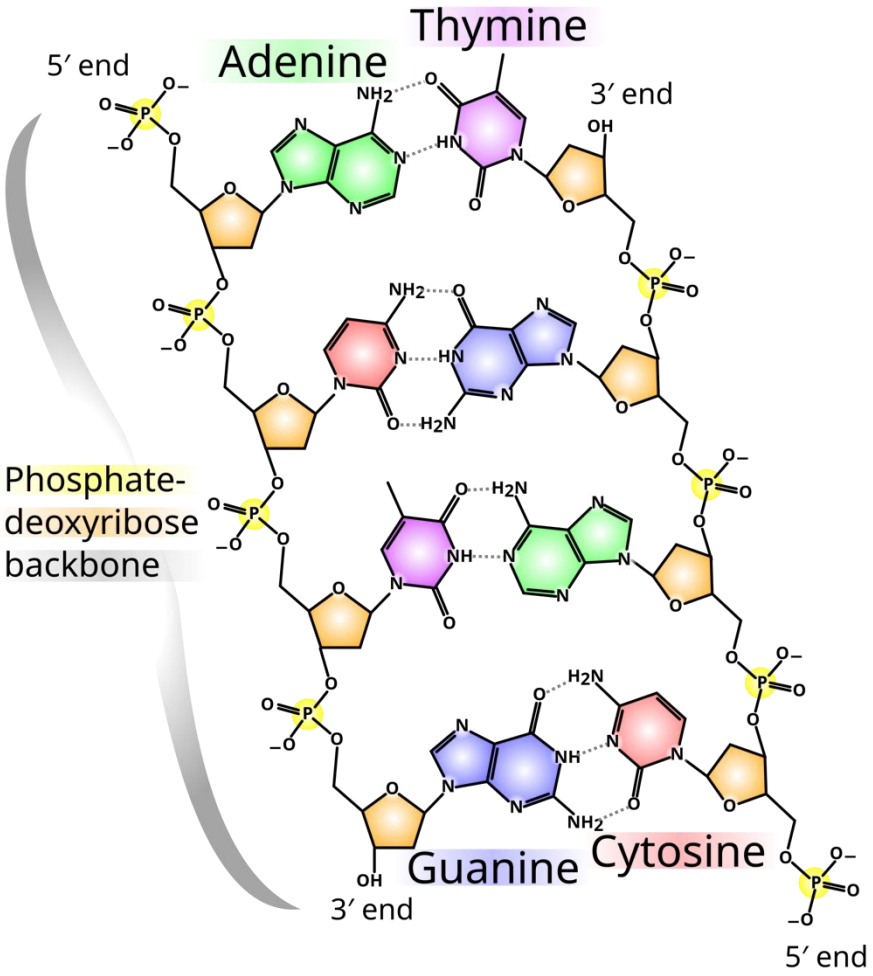
Learning outcomes

1) Describe the chemical structure of DNA and RNA
2) Describe canonical and non-canonical base pairing and secondary structures
3) Describe and use the two-state model for DNA hybridization to compute thermodynamic equilibria, melting temperatures and kinetic processes
4) Describe and use the Nearest-Neighbour Model for hybridization thermodynamics

Lecture outline

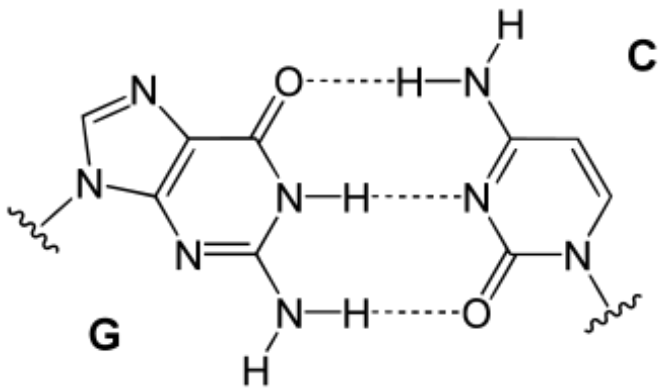
1) **Chemistry and structure of DNA and RNA**
2) Hybridisation thermodynamics
3) Hybridisation kinetics

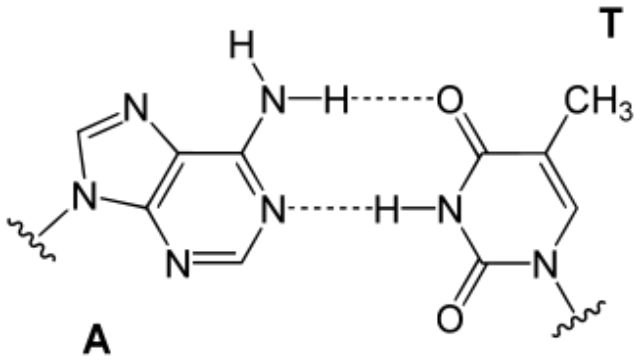
Structure of DNA



- Backbone consisting of a 5-carbon sugar (2-deoxyribose) and phosphate groups
- Sugar joined by phosphates between the 3’ and 5’ carbon atoms
- Oriented from 5’ to 3’ “ends” having a phosphate and hydroxyl group, respectively. **DNA is directional**
- Phosphate groups are negatively charged under philological pH and **repel each other**

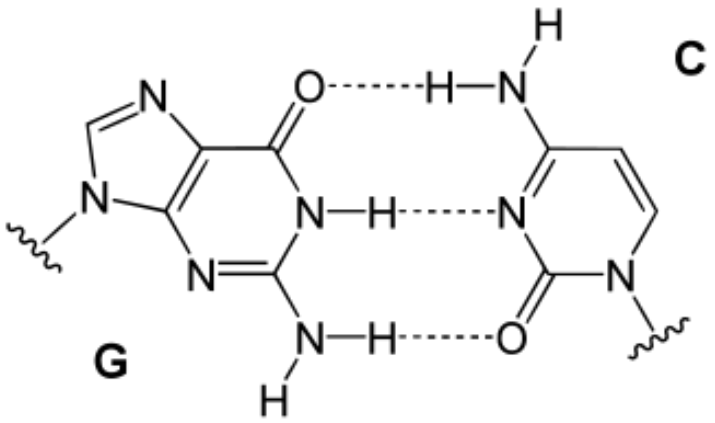
Structure of DNA

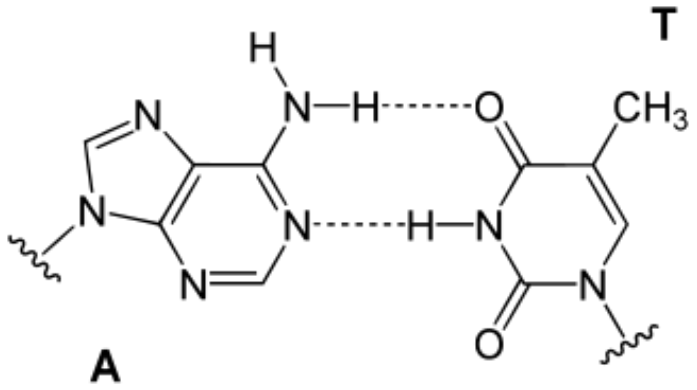




- Four **nucleobases** (or bases) can be connected to the phosphate-sugar backbone: **Guanine (G), Cytosine (C), Adenine (A) and Thymine (T)**
- G ad A are **purines**, with two rings
- C and T are **pyrimidines**, with one ring
- A base connected to the sugar is a **nucleoside**
- A base connected to the sugar and one (or more) phosphates is a **nucleotide** (nt), e.g. Adenosine Thiophosphate (ATP), Guanosine Monophosphate (GMP) etc.
- A single-stranded DNA molecule (or strand) is a **polynucleotide**
- If short (<100 nt), it is an **oligonucleotide**

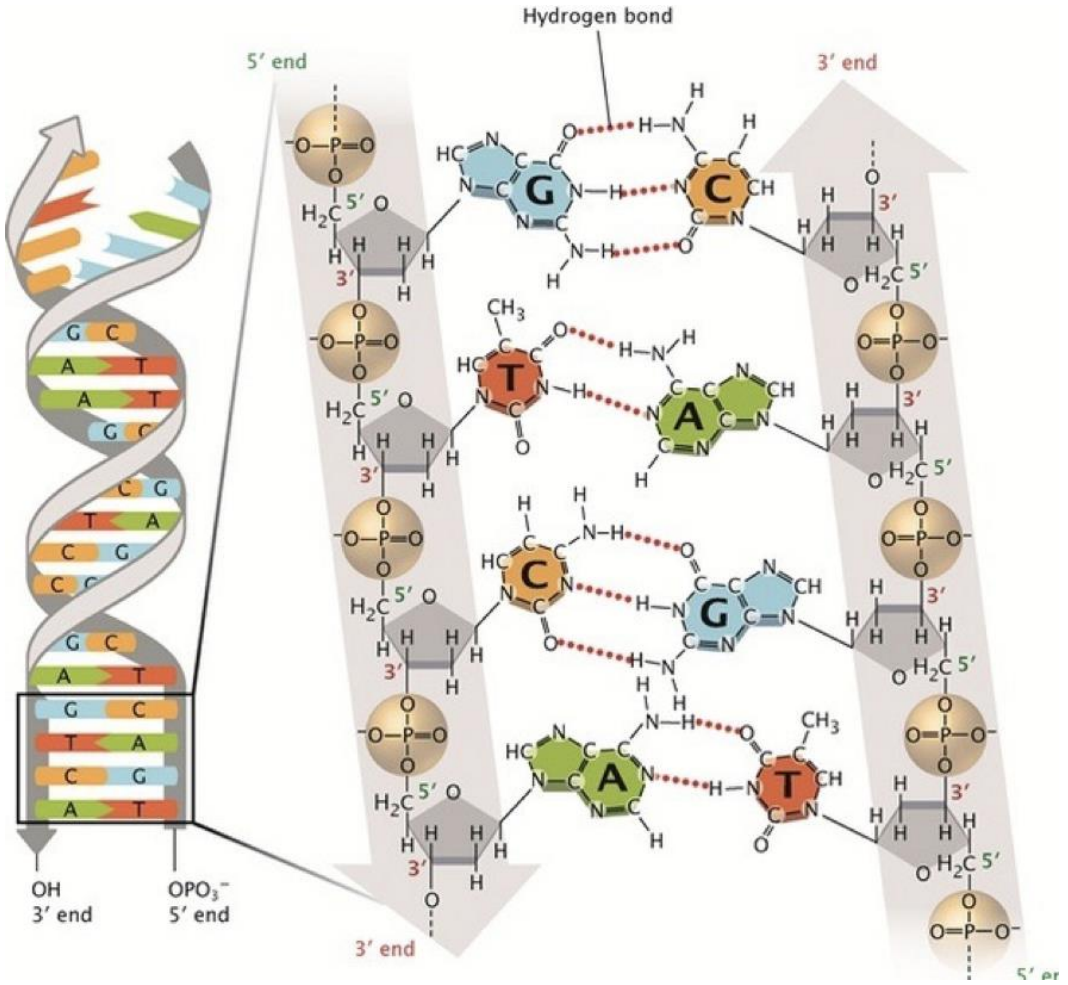
Canonical Base Pairing





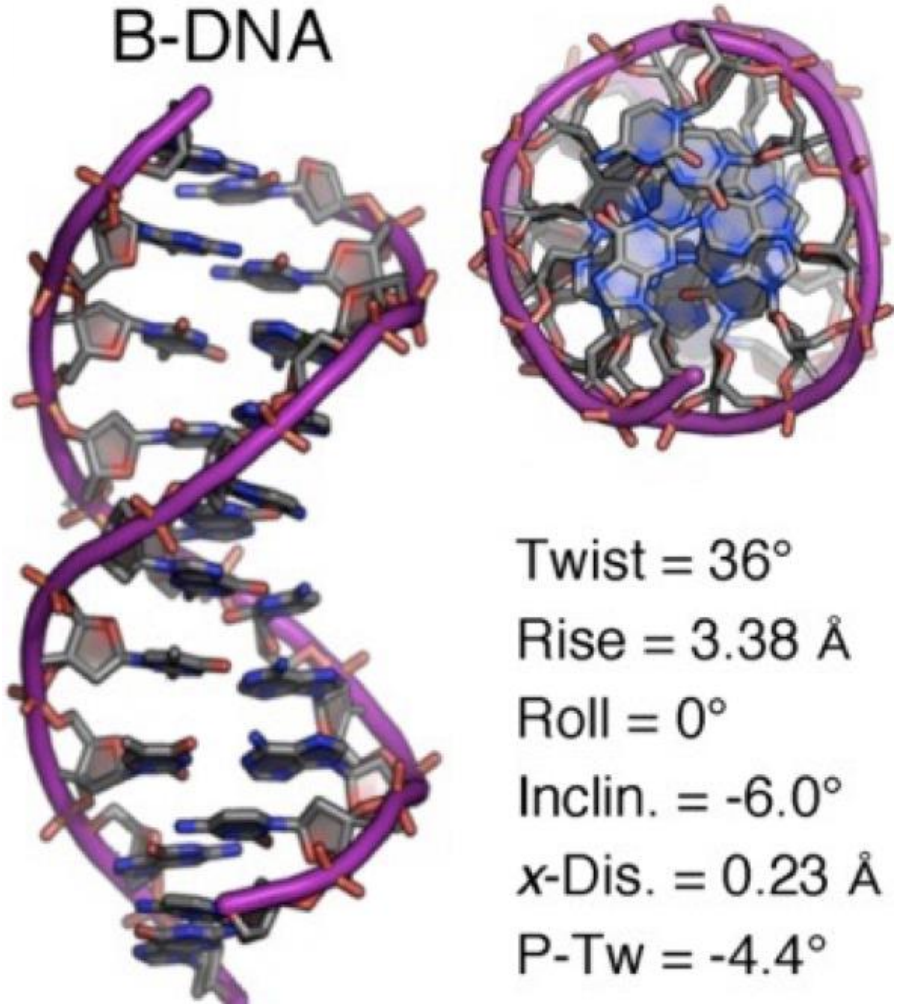
- Bases bind **selectively** in pairs: **A with T and C with G,** through canonical (Watson-Crick) base pairing
- Binding occurs via hydrogen bonding hence it is **thermally reversible** (very important for biological functions and nanotechnology)
- A and T form 2 hydrogen bonds (weaker bond), C and G form 3 (stronger bond)
- Bound nucleotides are referred to as **base-pairs** (bps)

DNA hybridisation



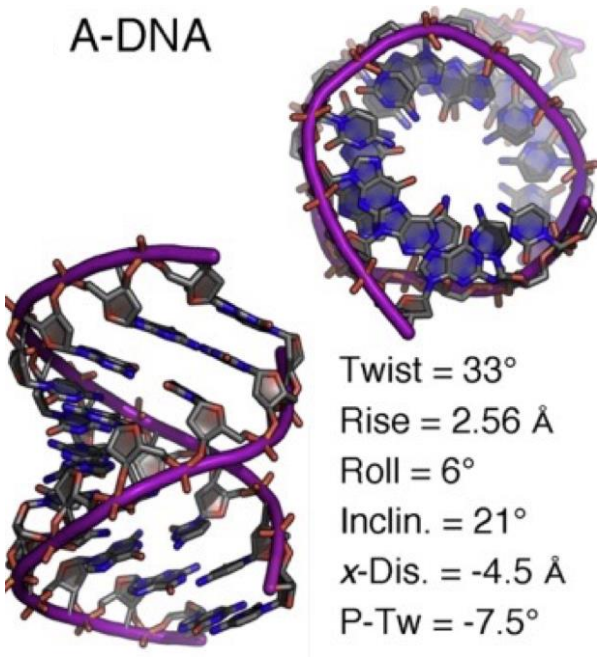
- Under suitable conditions, two **complementary** single-stranded DNA molecules can bind to form a double-stranded DNA **duplex**
- Strands bind with **anti-parallel directionality**
- Binding of two complementary DNA strands is called **hybridization**
- Some cations are required to screen phosphate charges and stabilize hybridisation

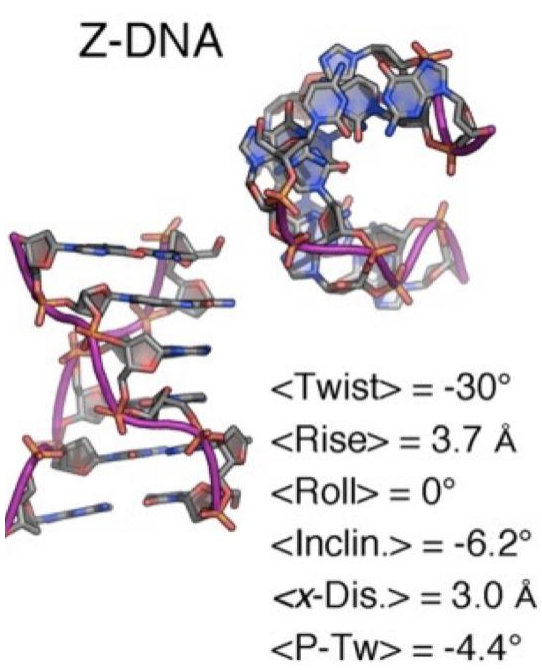
Double helix



- Under physiological conditions, double stranded DNA forms a **B-form double helix**
- The B DNA double helix is **right-handed**
- **Base stacking** interactions between nucleotides stabilise the double helix
- Double helix is asymmetric with a **minor and major groove**
- Structural properties:
  - Helical rise ~0.34 nm/bp
  - Width ~2 nm
  - ~10.5 bps (~3.6 nm) per helical turn

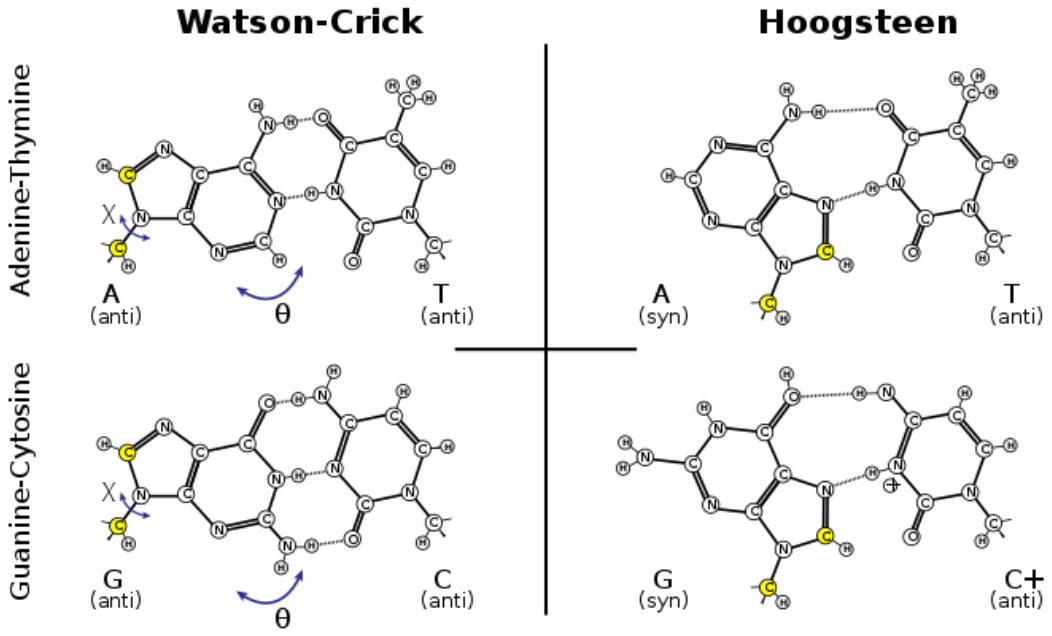
Double helices





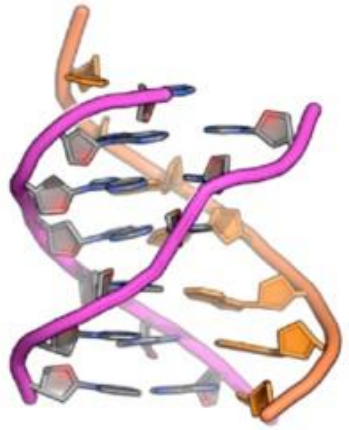
- Other forms of double-stranded DNA exist
- A-DNA, also a right-handed double helix forms rarely in DNA duplexes, but is relevant for DNA:RNA and RNA duplexes
- A-DNA is wider and more compressed
- Z-DNA is a left-handed double helix that can form in the presence of negative supercoiling

Hoogsteen base pairing



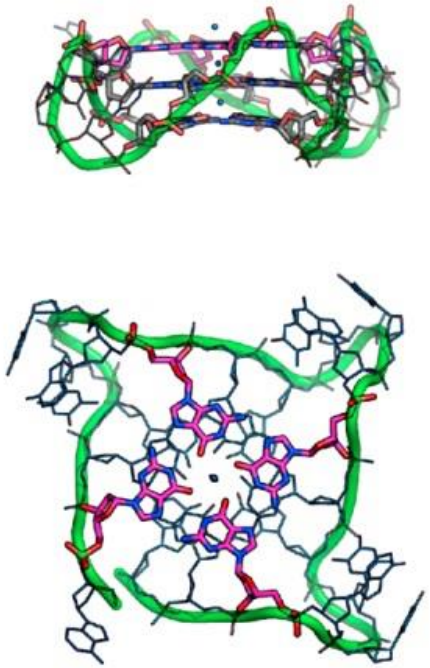
- A type of **non-canonical** (non Watson-Crick) base pairing
- Hydrogen bonds occur between the Watson-Crick (anti) face of a pyrimidine (e.g. T) an the syn face of a purine (e.g. A)
- **C needs to be protonated** to form Hoogsteen base pairing
- G can form similar hydrogen bonds with itself (syn vs anti face)

Non-canonical secondary structures



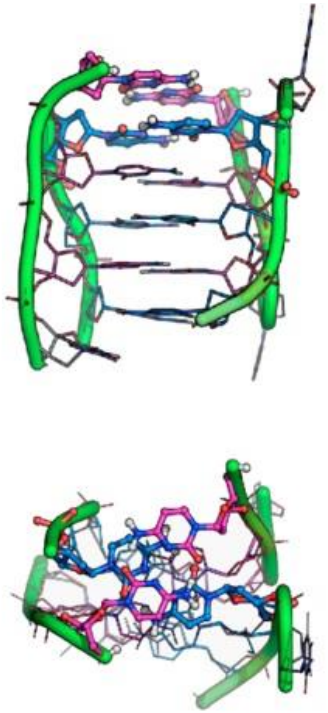
Triplex

- A thirds strand binds to a duplex (in the major groove) through Hoogsteen base pairing
- pH-dependent due to C protonation



G-quadruplex

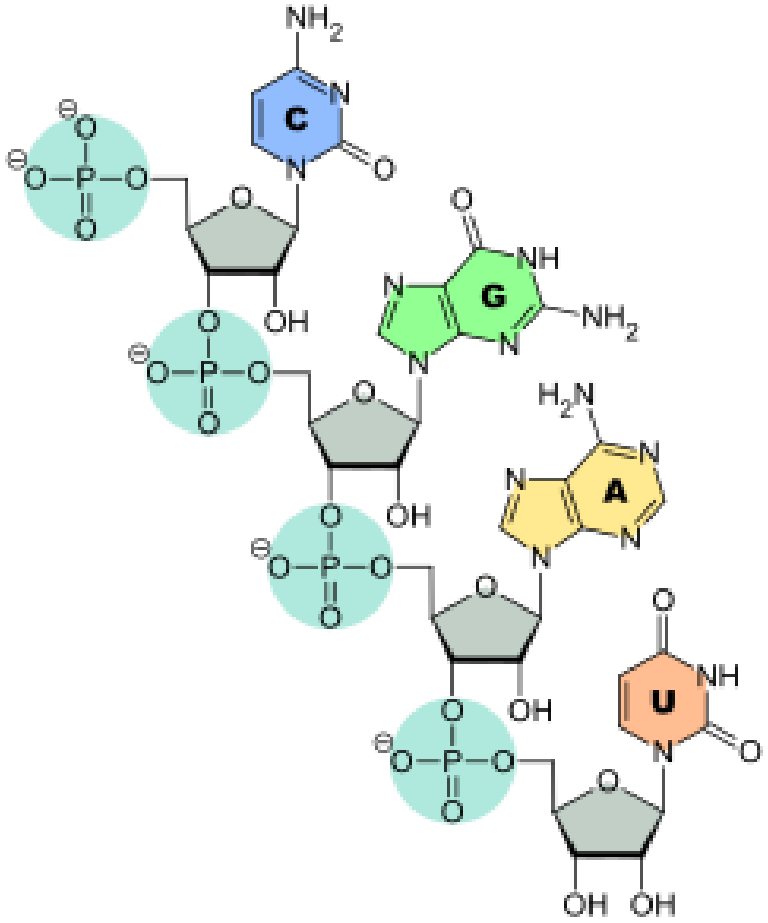
- G-rich sequences form compact "knot-like" structures of stacked **G-tetrads** held together by G-G Hoogsteen base pairing
- Coordinated by monovalent cations



i-motif

- Analogous to G-quadruplex, but for C-rich strands
- pH dependent due to C protonation

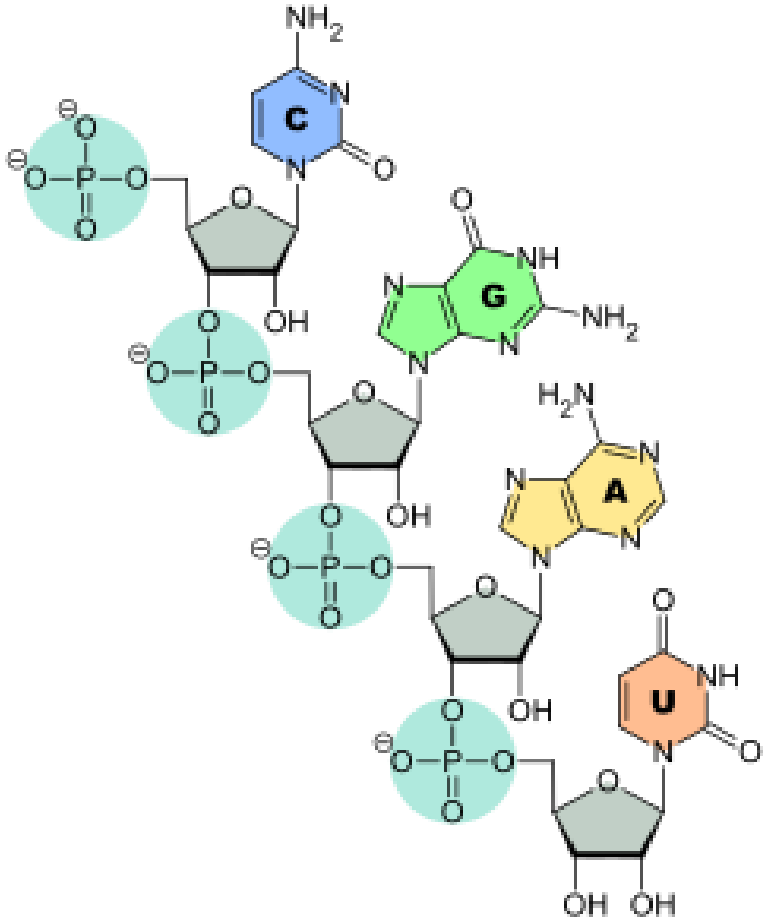
Structure of RNA



Same as DNA, but

- Sugar is **ribose**, with a hydroxyl group on the 2' position
- **Uracil** replaces thymine
- Uracil is the unmethylated form of thymine
- Uracil pairs with Adenine but can also form **non-canonical Wobble Pairs** with Guanine

Structure of RNA

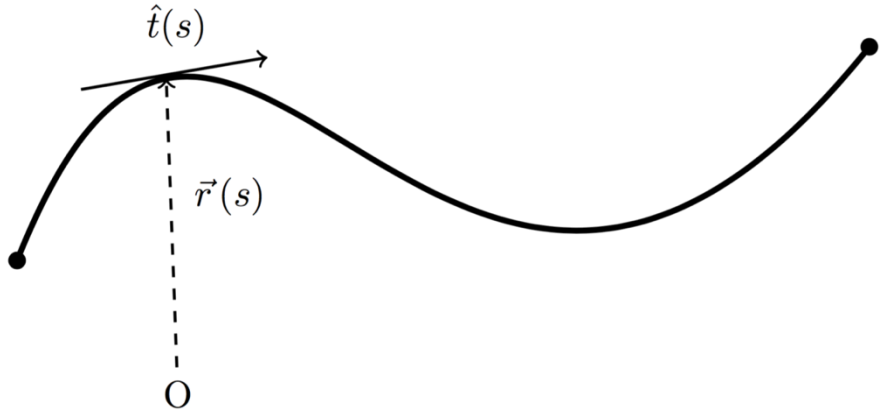


- Chemically more reactive due to (self)-**hydrolysis** induced by the OH group
- OH group also makes the molecule more flexible and capable of forming more complex secondary structures
- When double-stranded, RNA forms an **A-type double helix**

Polymer physics: worm-like chain model

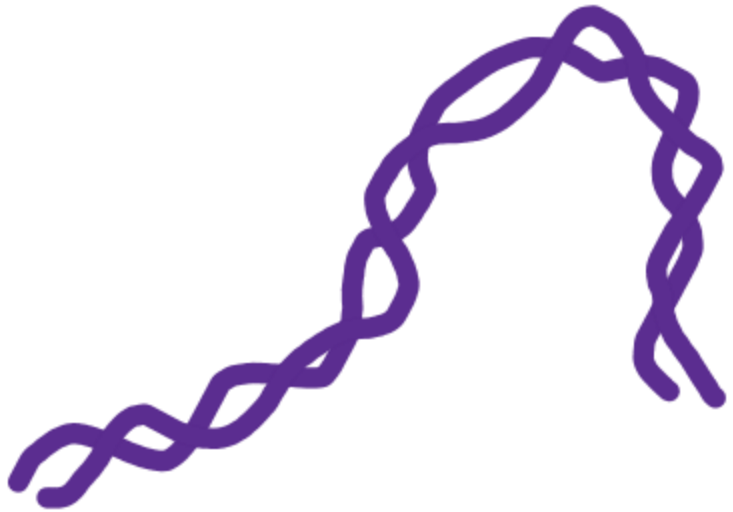
$$\langle \hat{t}(s)\hat{t}(0)\rangle = e^{-\frac{s}{l_P}}$$

l_P = persistence length. Distance over which the direction of the polymer loses correlation



Polymer physics of DNA





Single-stranded (ss) DNA

Highly flexible

$l_P \sim 1 - 2$ nm (3-6 nucleotides)

Free swivel

Double-stranded (ds) DNA

Much more rigid

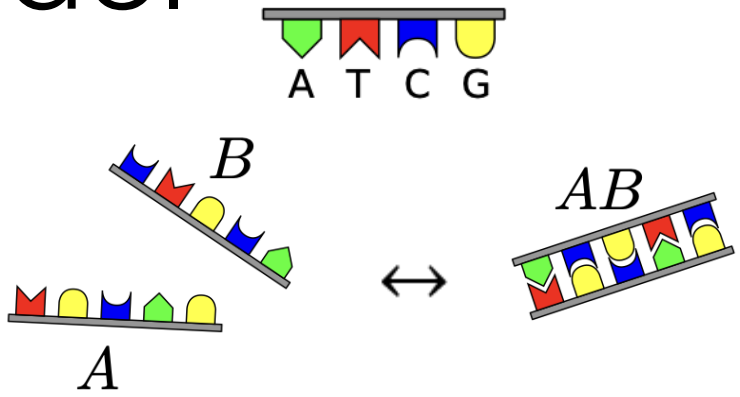
$l_P \sim 40 - 60$ nm (130-200 base-pairs)

Sustains torsion

Lecture outline

1) Chemistry and structure of DNA and RNA
2) **Hybridisation thermodynamics**
3) Hybridisation kinetics

Hybridisation thermodynamics: the two-state model



- Consider two complementary DNA **oligonucleotides**, A and B
- Assume that A and B are **either hybridised or separated**: partially bound states are neglected
- Describes (de)hybridisation as a reversible second order chemical reaction
- Equilibrium is determined by the Standard Gibbs Free Energy of Hybridisation, ΔG^0 (standard conditions, namely $\rho_0 = 1\mathrm{M}$)

$$A + B \rightleftharpoons AB$$

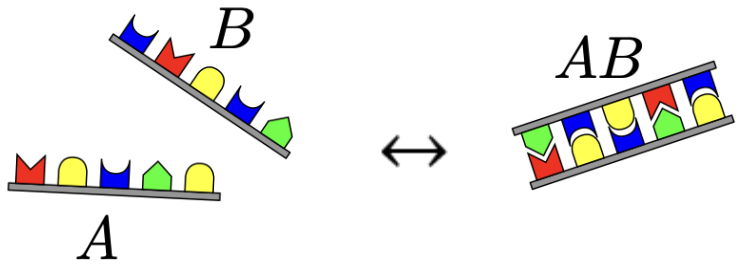
$$K = \frac{[AB]}{[A][B]} = \frac{1}{\rho_0} \exp\left(-\frac{\Delta G^0}{RT}\right)$$

$\rho_0 = 1\mathrm{M}$ concentration
K = equilibrium constant
ΔG^0 = Standard Gibbs free energy of hybridization
R = ideal gas constant
T = absolute temperature

Hybridisation Gibbs Free Energy

- ΔG^0 is defined as the **change in Gibbs Free Energy between the single-stranded and duplex states**
- It can be decomposed in enthalpic and entropic components: $\Delta G^0 = \Delta H^0 - T\Delta S^0$
- ΔH^0 = Standard hybridization enthalpy is negative (lower in duplexes) due to hydrogen bonding and (primarily) stacking. **It is the driving force for hybridization**
- ΔS^0 = Standard hybridization entropy, is also negative (lower in duplexes) due to loss of configurational freedom following hybridization. **Due to the – sign, it "opposes" hybridization**
- ΔH^0 and $-T\Delta S^0$ tend to be similar in magnitude under physiological conditions

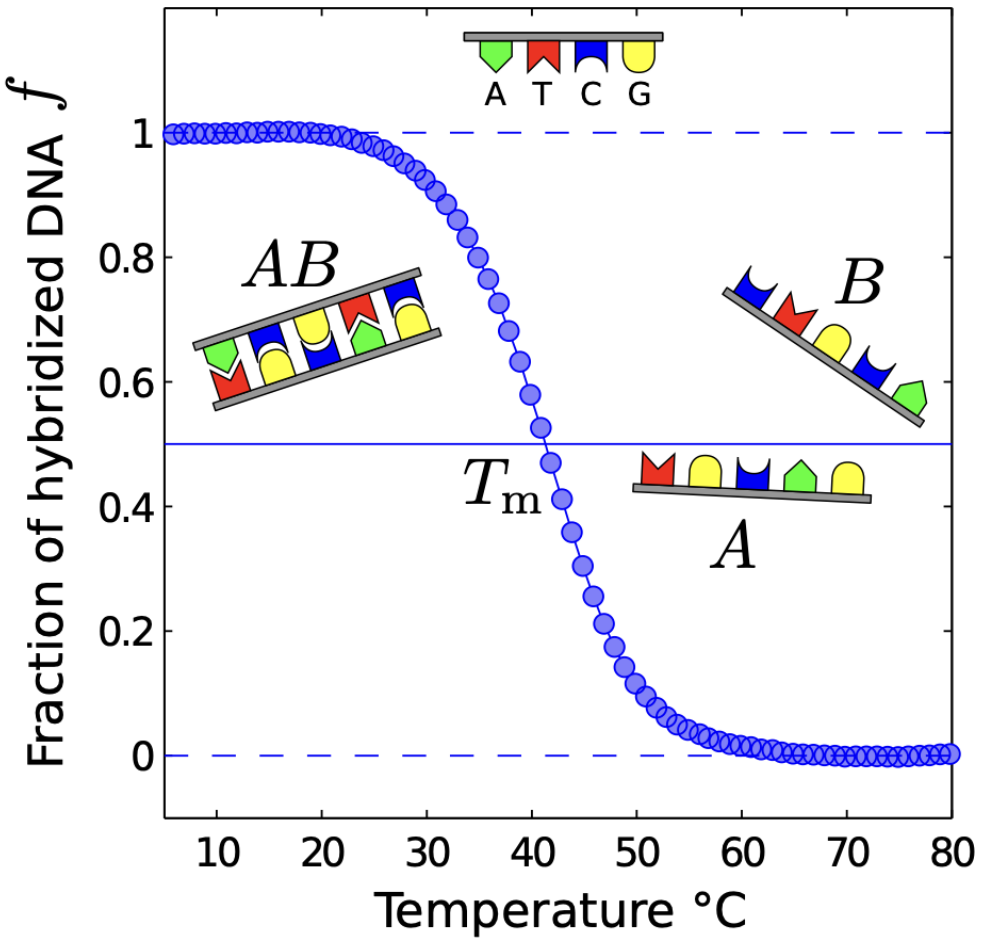




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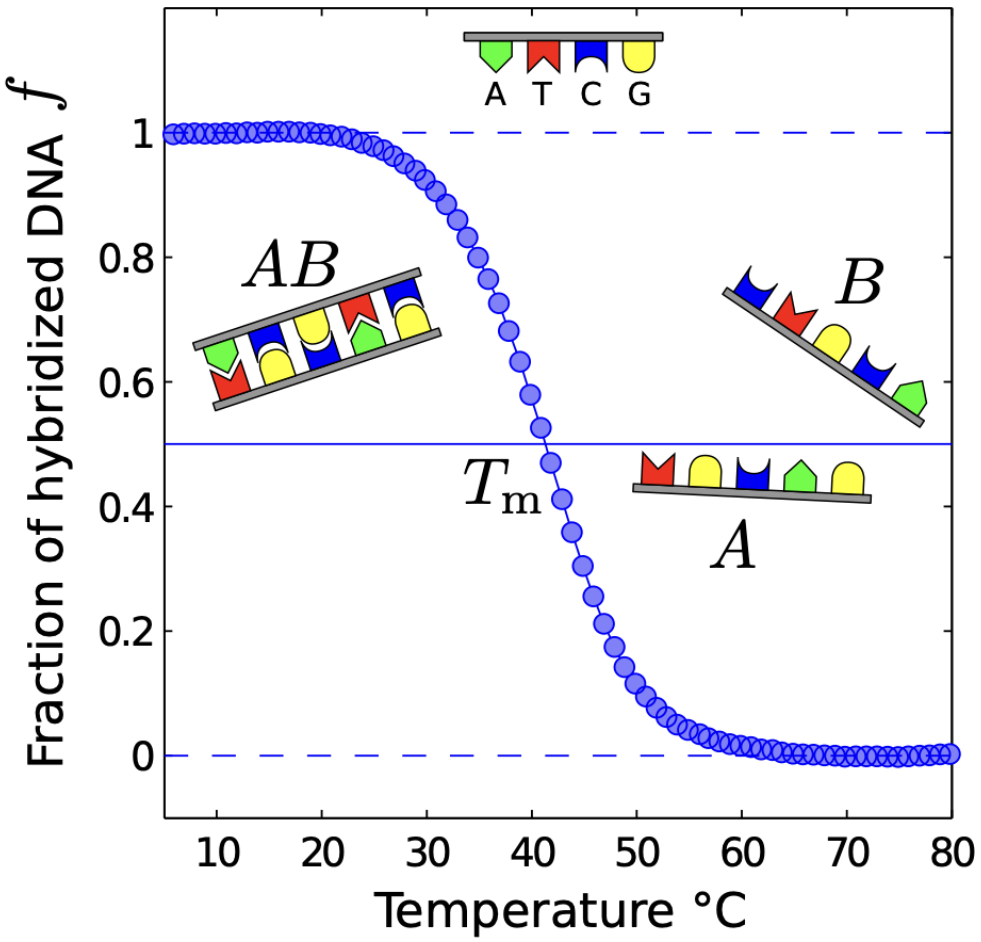
DNA melting (denaturation)



Experimental melting curve

- ΔH^0 and ΔS^0 are roughly temperature-independent
- $\Delta G^0 = \Delta H^0 - T\Delta S^0$ thus implies that DNA will denature if temperature is increased
- The temperature at which the probability of strands being hybridised is ½ is defined as the melting temperature, T_m

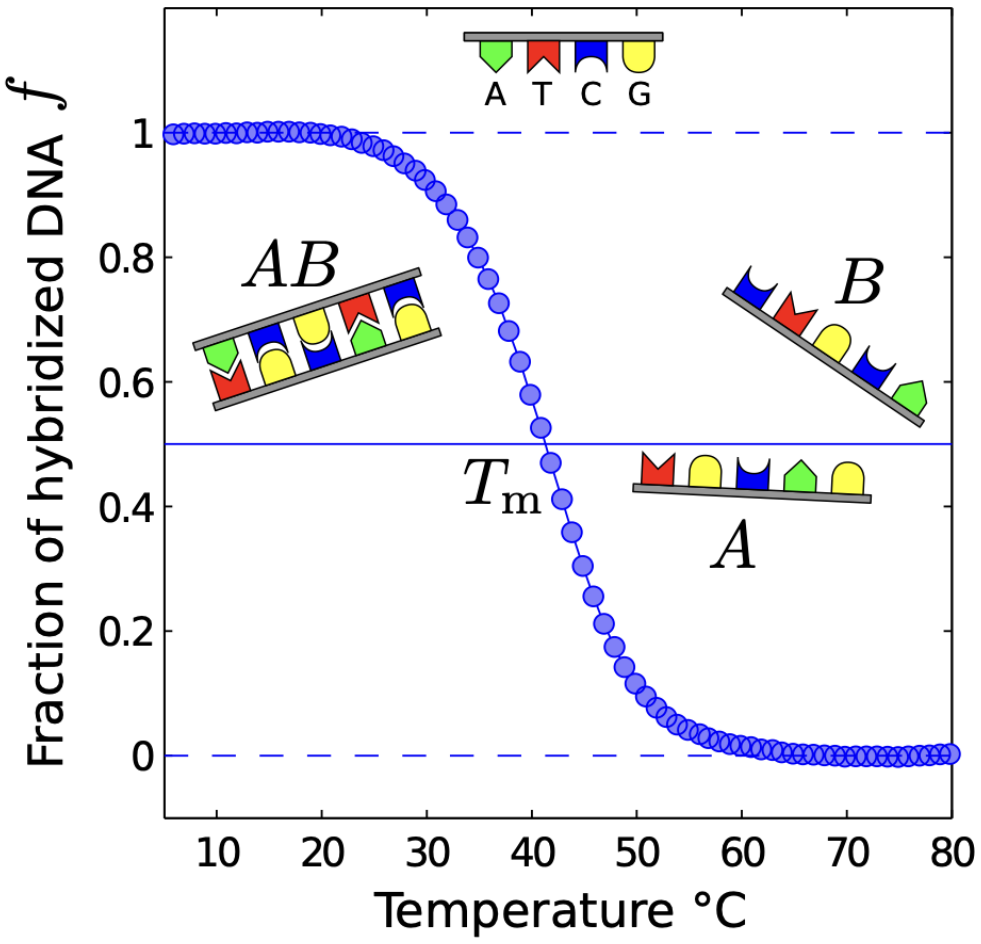
Exercise



Experimental melting curve

- Consider 2 complementary DNA oligos, A and B, which can hybridise to form duplex AB
- Assume that the total concentrations of the strands are $[A]_0 = [B]_0 = c$ and that the standard hybridisation enthalpy and entropy are ΔH^0 and ΔS^0
- Derive expressions for T_m

Solution



Experimental melting curve

We know $[A]_0 = [B]_0 = c$, hence $[AB] = fc$ and $[A] = [B] = c(1-f)$

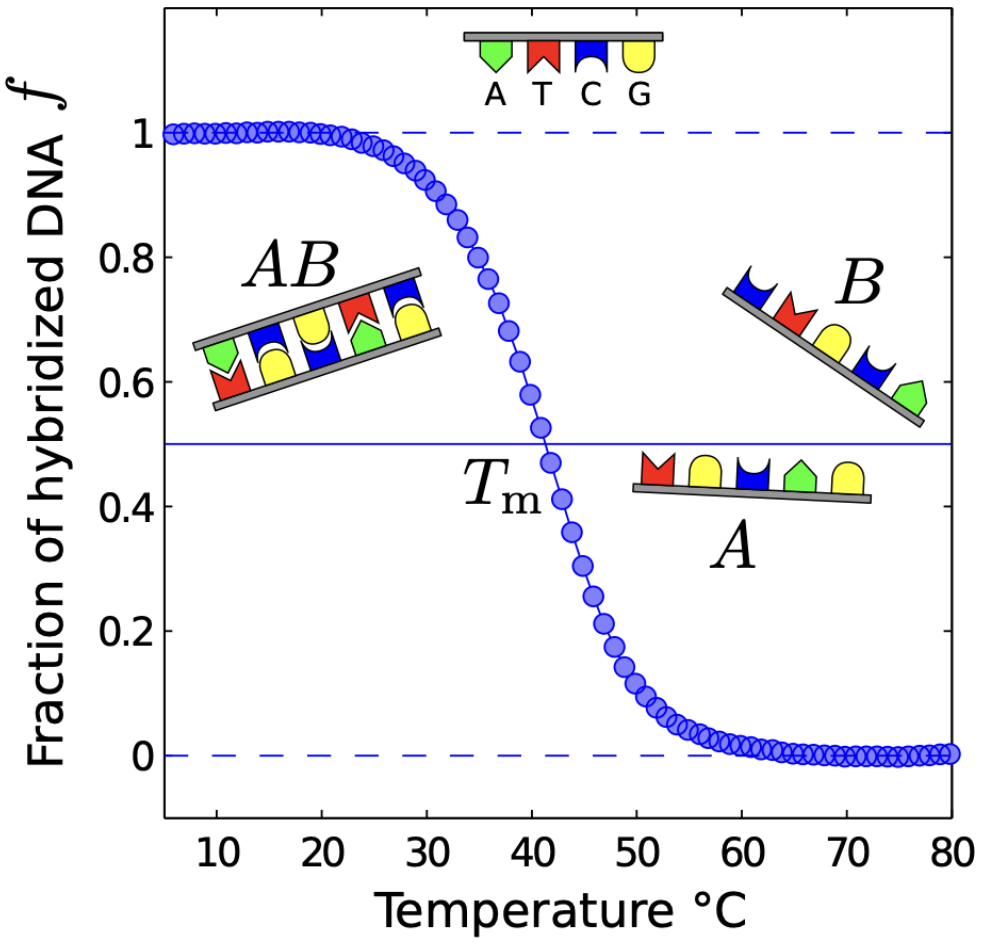
Substitute in

$$\frac{[AB]}{[A][B]} = \frac{1}{\rho_0}\exp\left(-\frac{\Delta G^0}{RT}\right)$$

to obtain

$$\frac{f}{c(1-f)^2} = \frac{1}{\rho_0}\exp\left(-\frac{\Delta H^0 - T\Delta S^0}{RT}\right)$$

Solution



Experimental melting curve

$$T = T_m \rightarrow f = \frac{1}{2}$$

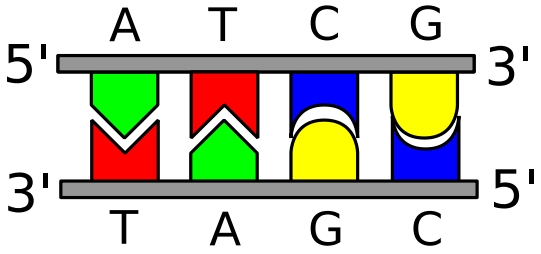
Hence

$$\frac{c}{2} = \rho_0 \exp\left(\frac{\Delta H^0}{RT_m} - \frac{\Delta S^0}{R}\right)$$

Rearranging

$$T_m = \frac{\Delta H^0}{\Delta S^0 + \mathrm{R} \ln\left(\frac{c}{2\rho_0}\right)}$$

The Nearest-Neighbours Thermodynamic Model



$$\Delta G^0 = \Delta G^0_{init,AT} + \Delta G^0_{AT/TA} + \Delta G^0_{TC/AG} + \Delta G^0_{CG/GC} + \Delta G^0_{init,GC}$$

- Enables estimation of ΔG^0 knowing the sequence
- ΔG^0 can be expressed as the sum of contributions associated to **neighbouring pairs of base-pairs** plus **initiation parameters**, associated to the formation of the first base-pair
- Why consider pairs of base pairs rather than adding up terms corresponding to individual base pairs? It is critical to consider neighbouring base-pairs due to the impact of **stacking interactions**

The Nearest-Neighbours Thermodynamic Model

There are 10 possible pairs of base-pairs (right), hence ΔG^0 can be written for an arbitrary sequence as

$$\Delta G^0 = \sum_{i=1}^{10} n_i \Delta G_i^0 + \Delta G_{init}^0 + \Delta G_{sym}^0$$

Where

- n_i is the number of instances that a given pair of base pairs (i) appears in a sequence
- $\Delta G_i^0 = \Delta H_i^0 - T\Delta S_i^0$ is the free energy contribution associated to base-pair-pair i
- ΔG_{sym}^0 is the symmetry correction, used for self-complementary (or palindromic) strands, e.g. ATCGAT

AA/TT
AT/TA
TA/AT
CA/GT
GT/CA
CT/GA
GA/CT
CG/GC
GC/CG
GG/CC

The Nearest-Neighbours Thermodynamic Model

- ΔH_i^0 and ΔS_i^0 have been determined experimentally from fitting and are tabulated
- Popular values from J. SantaLucia Jr, *PNAS*, 95(4) 1460-5 (1998) ("SantaLucia parameters")

Table 2. Unified oligonucleotide $\Delta H°$ and $\Delta S°$ NN parameters in 1 M NaCl (22)

Sequence	$\Delta H°$ kcal/mol	$\Delta S°$ cal/k·mol
AA/TT	−7.9	−22.2
AT/TA	−7.2	−20.4
TA/AT	−7.2	−21.3
CA/GT	−8.5	−22.7
GT/CA	−8.4	−22.4
CT/GA	−7.8	−21.0
GA/CT	−8.2	−22.2
CG/GC	−10.6	−27.2
GC/CG	−9.8	−24.4
GG/CC	−8.0	−19.9
Init. w/term. G·C	0.1	−2.8
Init. w/term. A·T	2.3	4.1
Symmetry correction	0	−1.4

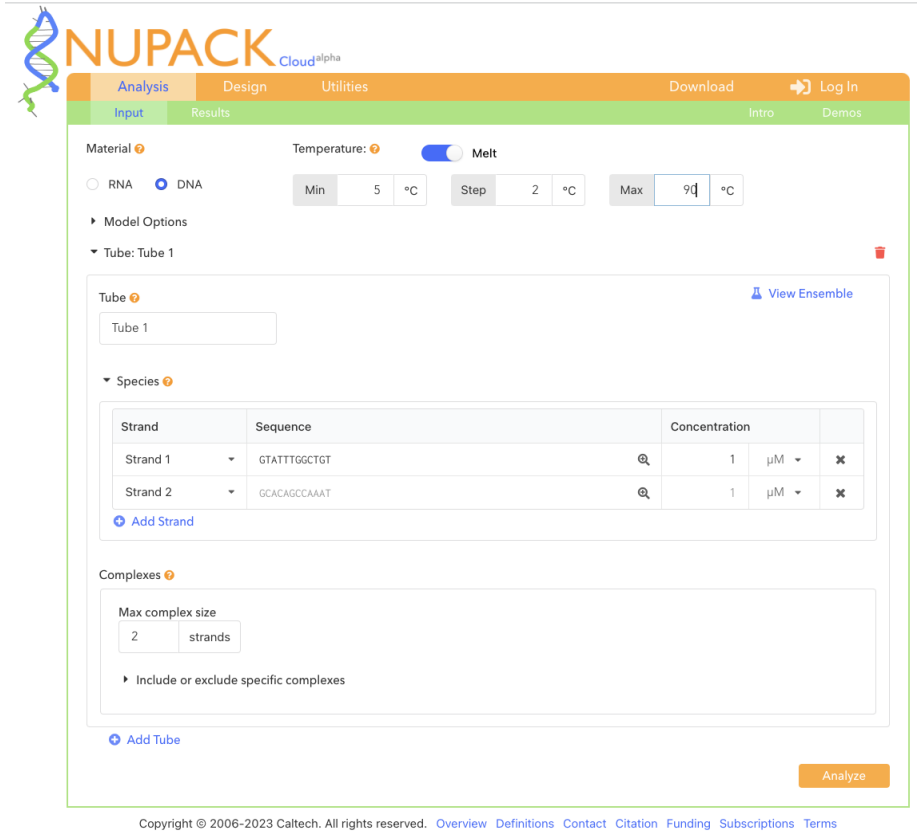
The Nearest-Neighbours Thermodynamic Model

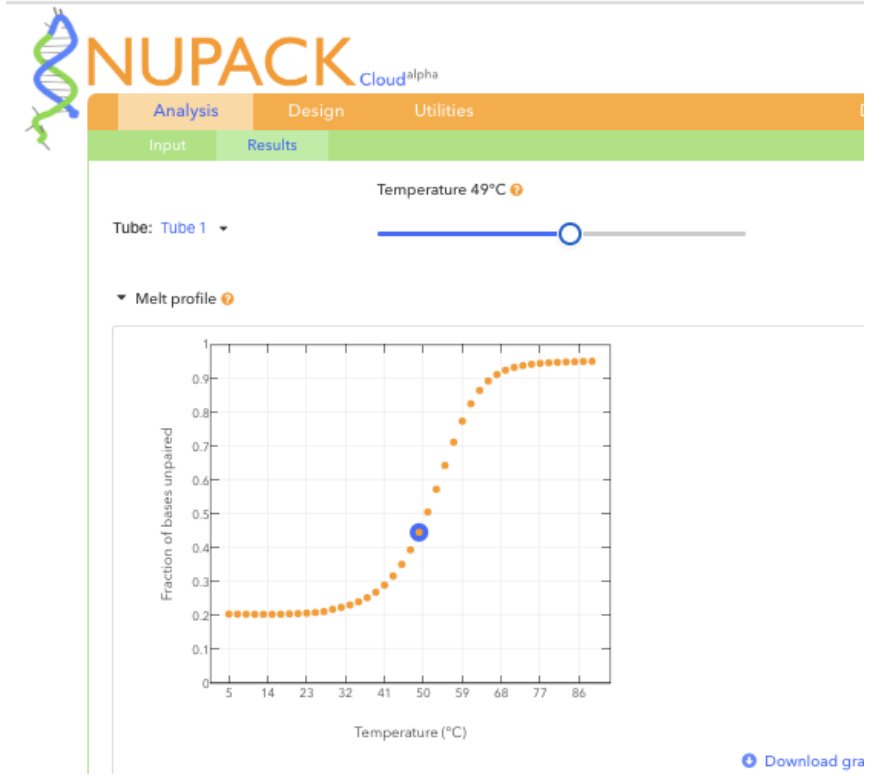
- ΔS_i^0 increases (logarithmically) with ionic strength of the solution and is different for monovalent (e.g. Na) vs divalent (e.g. Mg) cations
- ΔH_i^0 are salt-independent
- The symmetry correction, present for self-complementary (palindromic) strands, is purely entropic
- NN parameters are available for RNA:RNA and DNA:RNA base pairing, wobble pairs, mismatches etc.

Table 2. Unified oligonucleotide $\Delta H°$ and $\Delta S°$ NN parameters in 1 M NaCl (22)

Sequence	$\Delta H°$ kcal/mol	$\Delta S°$ cal/k·mol
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GT/CA	−8.4	−22.4
CT/GA	−7.8	−21.0
GA/CT	−8.2	−22.2
CG/GC	−10.6	−27.2
GC/CG	−9.8	−24.4
GG/CC	−8.0	−19.9
Init. w/term. G·C	0.1	−2.8
Init. w/term. A·T	2.3	4.1
Symmetry correction	0	−1.4

Computational tools: NUPACK (nupack.org)





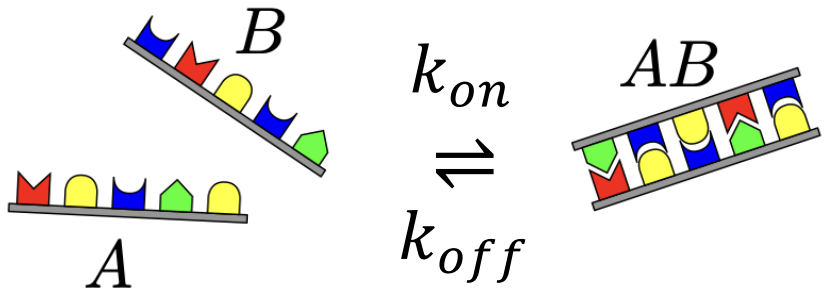
Lecture outline

1) Chemistry and structure of DNA and RNA
2) Hybridisation thermodynamics
3) **Hybridisation kinetics**

Hybridisation kinetics

$$A + B \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} AB$$

$$K = \frac{k_{on}}{k_{off}} = \frac{[AB]}{[A][B]} = \frac{1}{\rho_0} \exp\left(-\frac{\Delta G^0}{RT}\right)$$



k_{on}~$10^4 - 10^6$ $\mathrm{s^{-1}M^{-1}}$ depending on strand length, ionic strength, secondary structure…

$$k_{off} = k_{on}\, \rho_0 \exp\left(\frac{\Delta G^0}{RT}\right)$$

Assuming k_{on}~10^5 $\mathrm{s^{-1}M^{-1}}$

k_{off} (5 bp)~10^2 $\mathrm{s^{-1}}$
k_{off} (8 bp)~10^1 $\mathrm{s^{-1}}$
k_{off} (12 bp)~10^{-4} $\mathrm{s^{-1}}$
k_{off} (20 bp)~10^{-14} $\mathrm{s^{-1}}$
$1/k_{off}$ (20 bp)~10^7 years!